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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026
Fifth SEMESTER
Discipline Specific Core Course (DSC) - MG5DSCCHE300 – ORGANIC CHEMISTRY-3

Part A

Multiple Choice Questions

Answer **all questions**

Each question carries **1 marks**

1. **Which reagent selectively reduces only one nitro group in polynitro compounds?**
   d). NaSH
2. **What is the hybridization of the carbon atom in the aci-nitro form of nitromethane?**
   b).sp^2
3. **An alkyl halide reacts with KCN and AgCN separately. The products are respectively:**
   b). Nitrile and isocyanide
4. **The product obtained by reduction of CH_3CN with Stannous chloride /HCl followed by steam distillation is**
   c). CH_3CHO
5. **Benzene diazonium chloride reacts with CuBr to give:**
   a).Bromobenzene
6. **Nitration of benzoic acid predominantly gives:**
   c). m-Nitrobenzoic acid
7. **The reaction between sodium ethoxide and methyl iodide gives:**
   d). Ethyl methyl ether
8. **The pyrimidine base characteristic of RNA is**:
   c). Uracil
9. **Rhodopsin consists mainly of:**
   a). Retinal and opsin
10. **Paternò–Büchi reaction involves the reaction of a carbonyl compound with**
    b). Alkene

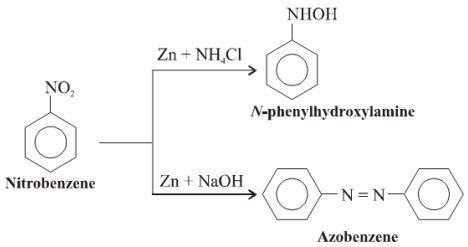
Part B

Short Answer Questions

Answer any **4 questions**

Each question carries **5 marks**

11. **Identify the major products and write the chemical equations for the reduction of nitrobenzene using:**
    a) Zn dust/NH_4Cl b) Zn dust/NaOH
    (a) Under mild condition, the reduction of nitrobenzene is controlled and stops at the hydroxylamine stage with the proper equation.
    (b) Under strong alkaline condition, intermediate reduction products (like nitroso benzene and phenylhydroxylamine) undergo bimolecular coupling/condensation, followed by further reduction to the final azo compound with proper equation.



(Correct product with equation give full marks 2.5+2.5)

12. Compare Gabriel's phthalimide synthesis and the Hoffmann bromamide reaction with respect to starting materials, reagents, mechanism, and products.

Sl No.	Gabriel's phthalimide synthesis	Hoffmann bromamide reaction
1	Used for synthesis of pure aliphatic primary amines.	Used for synthesis of both aliphatic and aromatic primary amines with one fewer carbon atom.
2	**Phthalimide** and an **alkyl halide** (must be primary or secondary aliphatic; aromatic halides do not work) were used as starting materials.	An **amide** (aliphatic or aromatic aliphatic/aromatic primary amide) is used as starting material.
3	A strong base (e.g.KOH or NaOH)), followed by a nucleophile for cleavage (typically hydrazine) is used as reagents.	A halogen and a strong base (typically Br_2 and NaOH or KOH).
4	Potassium phthalimide (nucleophile) and an N-alkylphthalimide is formed as intermediate.	Isocyanate (R-N=C=O), formed via a molecular rearrangement as intermediate.
5	Follows S_N2 substitution reaction mechanism.	Follows Hoffmann rearrangement reaction mechanism where intramolecular migration of an alkyl/aryl group to an electron-deficient nitrogen

(Any 4 relevant comparison points give full marks-2.5+2.5 each)

13. A compound has a five-membered ring containing one oxygen atom and two C=C bonds. Identify the compound and draw its structure. Explain why it is classified as an aromatic heterocycle.

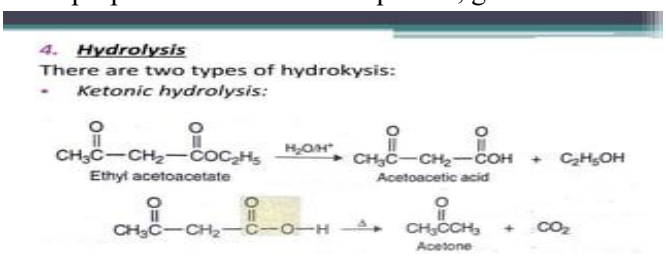
The compound is furan.

Furan is considered aromatic because it forms a planar, cyclic ring with a continuously conjugated system of six pi electrons, fulfilling Hückel's rule**.** One of the lone pairs on the oxygen atom resides in a p-orbital and delocalizes into the ring, completing the aromatic sextet alongside the four pi electrons from the two carbon-carbon double bonds.

(Correct name and structure: 1+1 marks and for correct explanation -3 marks)

14. Discuss the mechanism of ketonic hydrolysis of ethyl acetoacetate.

For a proper mechanism with equation, give full credit.



15. **Define electronic excitation and explain its significance in photochemistry.**

Definition: Electronic excitation is the process in which an electron absorbs energy and moves from a lower-energy electronic state to a higher-energy electronic state. This occurs when the absorbed energy matches the energy difference between the two electronic states.(Any proper definition give 2 marks).

Explanation: This process initiates photochemical reactions by generating highly reactive species, altering molecular geometry, and enabling pathways like fluorescence, phosphorescence, or energy transfer. (Any three relevant points give 3 marks)

16. **What is a Norrish Type I reaction? Explain with a suitable example.**

A Norrish Type I reaction is a photochemical process involving the homolytic cleavage of a carbonyl compound (aldehyde or ketone) at the bond between the carbonyl carbon and the alpha carbon atom (alpha-cleavage).

(Definition: 2 marks; Explanation with example: 3 marks)

Part C

Short Essay

Answer any **2 questions**

Each question carries **10 marks**

17. **Explain the complete procedure for identifying an unknown amine using the Hinsberg test. Include the observations and chemical reactions involved.**

Correct reagent identification and initial alkaline reaction-3 Marks

Clear differentiation mechanism of primary vs. secondary amines via acidification-4 Marks

Proper identification of tertiary amines and inclusion of all chemical equations-3 Marks

18. **Write a brief note on the structure of epoxides and the various reactions of Epoxides.**

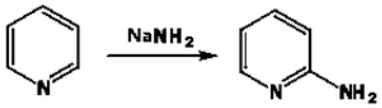
The structure section, requiring a diagram of the highly strained three-membered oxirane ring with an explanation of its bond angles and resulting ring strain.-Give 4 marks

The reactions section, expecting examples for either acid- catalysed or base-catalysed ring-opening reactions, or any other three reactions.-Give 6 marks

19. **a) Give one method of preparation of each of: i) Furan, iii) Pyrrole and iii)Thiophene.**
b) What happens when pyridine reacts with sodamide? Name the reaction.

(a) The method of preparation: Give 2 marks each (2+2+2);

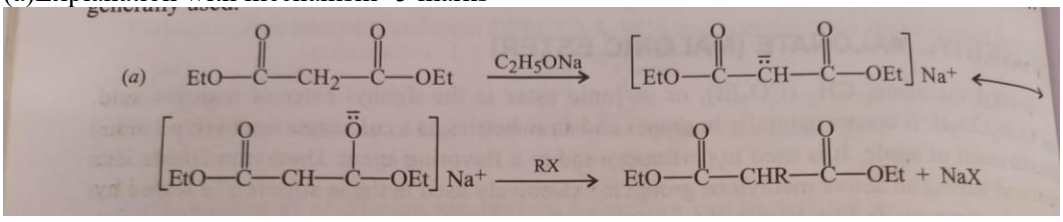
(b) Chichibabin reaction-Amination of pyridine



Give 3 marks for the reaction,1 mark for the name of the reaction.

20. **a) Explain the reaction of diethyl malonate with sodium ethoxide followed by an alkyl halide. Discuss the mechanism.**
b) Describe the synthesis of acetic acid using diethyl malonate.

(a)Explanation with mechanism- 5 marks



(b) Synthesis of acetic acid (Hydrolysis followed by decarboxylation of diethyl malonate gives acetic acid); correct synthesis equations - 5 marks